FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Advanced)-2020

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

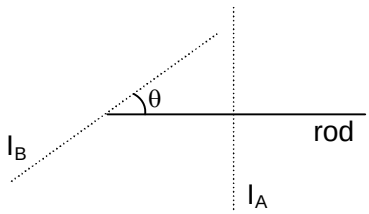
Physics

PART – I

SECTION – A

1. ABC

Sol. $I_B = \frac{m\ell^2}{3}\sin^2\theta,\ I_A = \frac{m\ell^2}{12}$



2. A

Sol. For open circuit, current = 0.

3. ABC

Sol. Assume x-axis along OA and y-axis perpendicular to it in the plane of ring.
After 1st collision

Velocity of particle = $\frac{v}{\sqrt{2}}$ along x-axis

And velocity of ring = $\frac{v}{\sqrt{2}}$ along y-axis

After second collision velocity of ring = $\frac{v}{\sqrt{2}}\hat{i} + \frac{v}{\sqrt{2}}\hat{j}$

Velocity of particle = 0

Time to return at point A = $4 \times \frac{R\sqrt{2}}{v}$

4. AB

Sol. $\dfrac{\dfrac{E}{2}+\dfrac{12}{4}+\dfrac{4}{6}}{\left(\dfrac{1}{2}+\dfrac{1}{4}+\dfrac{1}{6}\right)} = E - 2\times\dfrac{1}{2}$

$\Rightarrow$ E = 6.6, $\dfrac{12-(6.6+2\times0.5)}{4} = 1.1$

5. A

Sol. $V_B + 35\times10^{-3}\times(-10)^3 + 10 + 5\times5 = V_A$

$\Rightarrow$ $V_A - V_B = 0$.

6. ABCD

Sol. L.C. = pitch / number of division on circular scale. The result has to be expressed upto least number of significant figures.

7. B

Sol. $E_{3500\,\mathring{A}^0} = \dfrac{12420}{3500}\text{eV} = 3.55\text{ eV}$

$\therefore \quad \phi_B < E_{3500\,\mathring{A}^0} < \phi_A$

8. AC

Sol. $\omega_L = 50\ \Omega$

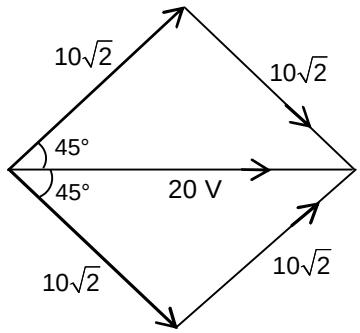
$X_C = \dfrac{1}{\omega_C} = 100\ \Omega$

$i_C = \dfrac{10\sqrt{2}}{100} = \dfrac{\sqrt{2}}{10} \quad ; \quad i_L = \dfrac{10\sqrt{2}}{50} = \dfrac{\sqrt{2}}{5}$

$i = \sqrt{i_C^2 + i_L^2}$

$= \dfrac{\sqrt{2}}{10}\sqrt{1+4} = \dfrac{\sqrt{10}}{10} = 0.316$ A

$V_{100\,\Omega} = 10\sqrt{2}$



9. A

Sol. O to A the slope is positive.
A to B the slope is negative.
B to C the slope is positive.
At A the slope is zero

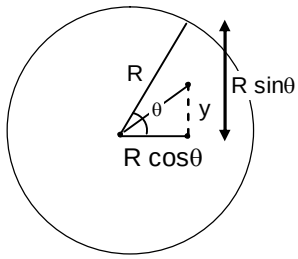
10. B

Sol. O to A the graph is convex.
A to B the graph is first convex and then concave.
B to C the graph is concave.
At point A the graph is convex.

11. A

12. A

Sol. (for Q.11 & Q.12)



$$N = \frac{4\pi G\rho}{3} R\cos\theta$$

$$a = -\frac{4\pi G\rho}{3} y$$

$$|a_{max}| = \frac{4\pi G\rho}{3} R\sin\theta$$

SECTION – C

13. 00007.00

Sol. $$\frac{P^2}{2m} = KE = \frac{9\times10^9\times120e^2}{10\times10^{-15}}$$

$$\lambda = \frac{h}{P} = \frac{h}{e\sqrt{360\times10^{-3}}} = \frac{4.2\times10^{-15}}{0.6} = 7.$$

14. 00004.00

Sol. $$300 = f = \frac{1}{2L}\sqrt{\frac{20g}{\mu}}, f' = \frac{1}{2L}\sqrt{\frac{4\times5\times15g}{(5+15)\mu}} \Rightarrow \frac{f'}{300} = \sqrt{\frac{15}{20}} \Rightarrow f' = 259.80 \text{ Hz}$$

15. 00000.80

Sol. $$\delta = \left(\frac{\mu_2}{\mu_1} - 1\right)A = \left(\frac{1.5}{5/4} - 1\right)\times4^0 = 0.8^0 \text{ downward}$$

16. 00001.00

Sol. $$h = u_y t - \frac{1}{2}gt^2$$

$$\Rightarrow\ t_1 + t_2 = \frac{-u_y}{-\frac{1}{2}g} = \frac{2u_T}{g} = \frac{2u\sin\theta}{g}$$

17. 00002.00

Sol. $\vec{F}_3 + \vec{F}_1 = \vec{F}_2$ (triangle law), so resultant force $= 2\vec{F}_2$.

$$\vec{a} = \frac{2\vec{F}_2}{m}$$

18. 00000.25

Sol. Applying conservation of momentum $mv = \frac{m}{2}\times0 + \frac{m}{2}v' \ \Rightarrow\ v' = 2v$

Increase in mechanical energy of the system $= \frac{1}{2}\frac{m}{2}(2v)^2 - \frac{1}{2}mv^2 = \frac{1}{2}mv^2 = \frac{GMm}{2\times(2R)}$

$$= \frac{mgR^2}{4R} = \frac{mgR}{4}$$

Chemistry

PART – II

SECTION – A

19. ACD

Sol. $\ln [A]_t = \ln [A]_0 - kt$ (for 1st order reaction)

slope of $\ln [A]_t$ vs t = -k

$k = 0.0231\ \text{sec}^{-1}$

$t_{1/2} = \frac{0.693}{0.0231} = 30\ \text{sec}$

Half life is independent of initial concentration. So a plot of $t_{1/2}$ vs concentration will give a straight line parallel to x – axis.

$t_{90\%} = \frac{1}{0.0231} \ln \frac{100}{10} = 100\ \text{sec}$

At higher temperature, value of rate constant k increases. So, the slope of the plot $\ln[A]_t$ vs t will increase and the line will become steeper.

20. AC

Sol. $SO_2 > CH_4$ (depends on van der Waal's force)

$CO > N_2$ (CO is polar and N_2 is non-polar)

21. C

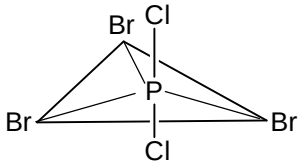
Sol. Charge = Area of (I vs t) curve

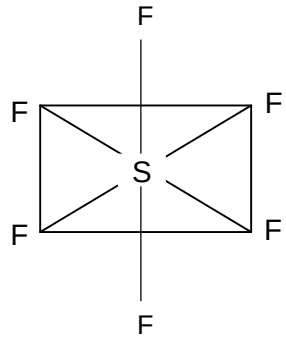
$= \frac{1}{2} \times 10 \times (100 \times 10^{-3}) + 10 \times (100 \times 10^{-3}) + \frac{1}{2} \times 10 \times (100 \times 10^{-3})$

$= \frac{1}{2} + 1 + \frac{1}{2} = 2$ coulomb

22. BD

Sol. Both the molecules have symmetrical structure.





23. AC

Sol. $NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$

$Fe^{2+} + H_2O \rightleftharpoons \left[Fe(OH)\right]^+ + H^+$

The H^+ ions combine with or remove OH^- ions by forming water and drives the reaction towards forward direction.

24. ABD

Sol. (C) cannot be prepared by hydrolysis of any alkene.

25. ABC

Sol. $RCOONa \longrightarrow RCOO^- + Na^+$

$RCOO^- \longrightarrow R^\bullet + CO_2$

$CH_3\overset{\bullet}{C}H_2 + CH_3\overset{\bullet}{C}H_2 \longrightarrow CH_3CH_2CH_2CH_3$

$CH_3\overset{\bullet}{C}H_2 + \overset{\bullet}{C}H_3 \longrightarrow CH_3CH_2CH_3$

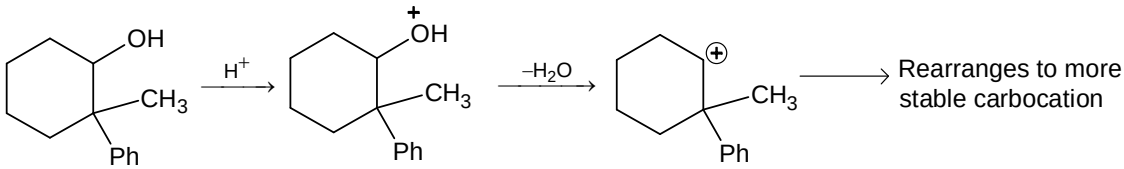
$\overset{\bullet}{C}H_3 + \overset{\bullet}{C}H_3 \longrightarrow C_2H_6$

26. ABCD

Sol. All the statements are correct.

27. A

Sol.

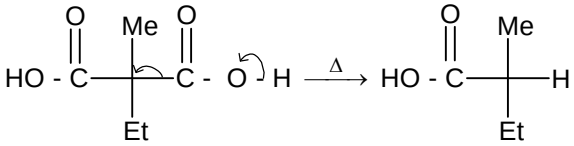


28. C

Sol. Self explained

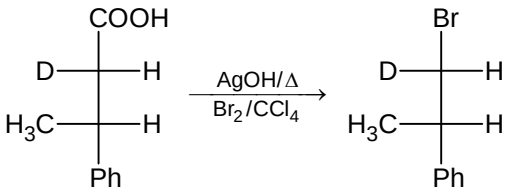
29. C

Sol.



30. B

Sol.



SECTION – C

31. 00002.00

Sol. Nitrogen and bismuth do not form pentachloride. In case of 'N', no vacant orbitals are present and for 'Bi', inert-pair effect is most prominent.

32. 00004.00

Sol. 1F deposits 1 eq or ½ mole of Cu

$\therefore$ 8F will deposit 4 mole of Cu

33. 00005.00

Sol. $\lambda = \frac{1}{t_{avg}} = \frac{1}{t}\ln\frac{[A_0]}{[A_t]}; \frac{1}{7.2} = \frac{1}{t_1}\ln\frac{100}{100-\frac{100}{3}}$

$$\frac{1}{7.2} = \frac{1}{t_2}\ln\frac{100}{100-\frac{200}{3}}$$

$t_2 - t_1 = 7.2\ \ln 3 - 7.2\ln\frac{3}{2} = 5$

34. 00000.46

Sol. $_{83}Bi^{211} \longrightarrow {}_{81}Tl^{207} + {}_2He^4$; total time = n × half-life

Moles of substance left after n halves = $\frac{\text{initial moles}}{2^n} = \frac{2}{2^4} = 0.125$

Moles of He produced = 2 – 0.125 = 1.875

Pressure developed due to

$He = \frac{1.875 \times 0.0821 \times 300}{100} = 0.4618$ atm

35. 00001.55

Sol. $1.50 = E^o - \frac{0.0591}{3}\log\left[\frac{0.3}{(0.1)^3}\right]$

$\therefore E^o = 1.55$

36. 00001.12

Sol. $E^o_{Cell} = \frac{0.0591}{2}\log K_{eq}$

$$0.003 = \frac{0.0591}{2}\log\frac{[D^+]^2}{[H^+]^2} \Rightarrow \frac{[D^+]}{[H^+]} = 1.123$$

Mathematics

PART – III

SECTION – A

37. AB

Sol. Let the length of the sides of the rhombus be l

$\Rightarrow$ Area $= l^2 \sin 30° = 2 \Rightarrow l = 2$ unit

$OB^2 = OA^2 + AB^2 - 2OA.AB\cos 150°$

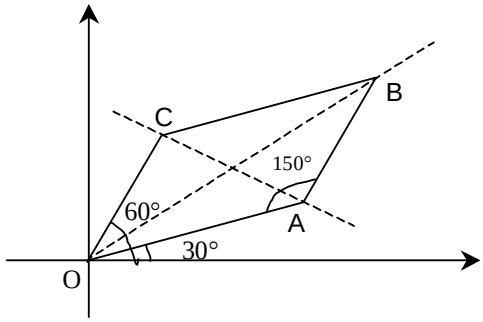
$= 4 + 4 - 2(4).\left(-\frac{\sqrt{3}}{2}\right) = 4(2 + \sqrt{3})$

$OB = 2\sqrt{2+\sqrt{3}}$

$B \equiv \left(\sqrt{4+2\sqrt{3}},\ \sqrt{4+2\sqrt{3}}\right) \equiv \left(1+\sqrt{3},\ 1+\sqrt{3}\right)$

Hence coordinates of B can be $\left(1+\sqrt{3},\ 1+\sqrt{3}\right)$

or $\left(-1-\sqrt{3},\ -1-\sqrt{3}\right)$



38. ACD

Sol. $z^2 + az + a^2 = 0$

$\Rightarrow$ $z = a\omega,\ a\omega^2$ (where 'ω' is non real root of cube unity)

$\Rightarrow$ locus of z is a pair of straight lines

and arg (z) =arg(a) + arg(ω) or arg(a) + arg(ω^2)

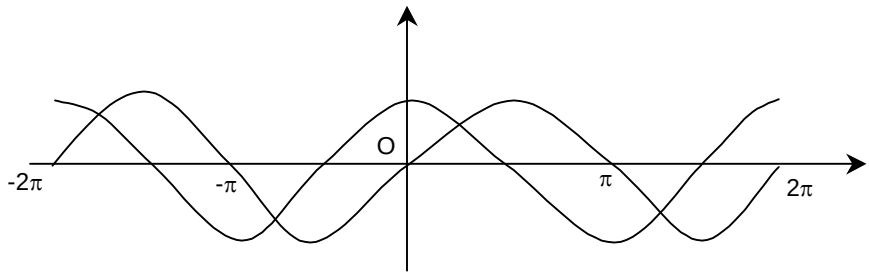
$\Rightarrow \arg(z) = \pm\frac{2\pi}{3}$

also, $|z| = |a||\omega|$ or $|a|\,|\omega^2| \Rightarrow |z| = |a|$

39. ABCD

Sol. $f'(x) = \frac{\cos x - \sin x}{1+(\sin x + \cos x)^2} > 0 \Rightarrow \cos x > \sin x$

$\Rightarrow x \in \left(-2\pi, -\frac{7\pi}{4}\right) \cup \left(-\frac{3\pi}{4}, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right)$



40. AB

Sol. $3x - 6y + 2z + 5 = 0$

$-4x + 12y - 3z + 3 = 0$

Bisectors are $\frac{3x-6y+2z+5}{\sqrt{9+36+4}} = \pm\frac{-4x+12y-3z+3}{\sqrt{16+144+9}}$

The plane which bisects the angle between the planes that contains the origin.

$13(3x-6y+2z+5)=7(-4x+12y-3z+3)$

$67x-162y+47z+44=0$

Further, $3x(-4)+(-6)(12)+2\times(-3)<0$

Hence, the origin lies in the acute angle.

41. ACD

Sol. $P(E)=\dfrac{^{2n}C_n}{2^{2n}}=\dfrac{(2n)!}{n!n!2^n\,2^n}$

$=\dfrac{1\times2\times3\times....x(2n)}{n!n!2^n\,2^n}$

$=\dfrac{1\times3\times5\times....\times(2n-1)}{n!2^n}$

Now,

$\prod_{r=1}^{n}\left(\dfrac{2r-1}{2r}\right)=\dfrac{1\times3\times5\times....\times(2n-1)}{2\times4\times6\times.....\times(2n)}$

$=\dfrac{1\times3\times5\times.....\times(2n-1)}{(1\times2\times3\times....\times n)2^n}$

$\sum_{r=0}^{n}\left(\dfrac{^nC_r}{2^n}\right)^2=\dfrac{1}{2^n\,2^n}\sum_{r=0}^{n}\left(^nC_r\right)^2$

$=\dfrac{1}{2^n2^n}\,^{2n}C_n$

Also,

$\dfrac{\sum_{r=0}^{n}\left(^nC_r\right)^2}{\left(\sum_{r=0}^{2n}{}^{2n}C_r\right)}=\dfrac{^{2n}C_n}{2^{2n}}$

42. AC

Sol. $f(n)=\begin{vmatrix} n & n+1 & n+2 \\ n! & (n+1)! & (n+2)! \\ 1 & 1 & 1 \end{vmatrix}=\begin{vmatrix} n & 1 & 1 \\ n! & nn! & (n+1)(n+1)! \\ 1 & 0 & 0 \end{vmatrix}$

[Applying $C_3\to C_3-C_2$ and $C_2\to C_2-C_1$]

$=(n+1)(n+1)!-nn!=n!\left[(n+1)^2-n\right]=n!=n!(n^2+n+1)$

Thus, $f(n)$ is divisible by n! and n^2+n+1

43. AD

Sol. $I=\dfrac{\sqrt{(1+\sin x)(1-\sin x)}}{\sqrt{\sin x(1-\sin x)}}=\dfrac{\cos x}{\sqrt{\sin x(1-\sin x)}}=\dfrac{\cos x}{\sqrt{\dfrac{1}{4}-\left(\dfrac{1}{2}-\sin x\right)^2}}=\int\dfrac{-dt}{\sqrt{\left(\dfrac{1}{2}\right)^2-t^2}}$

44. ABD

Sol. $A^2 - 4A - 5I_3 = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$

Multiplying by A^{-1}, $A - 4I - 5A^{-1} = 0 \Rightarrow A^{-1} = \frac{1}{5}(A - 4I)$

$|A| \neq 0 \Rightarrow |A|^2 \neq 0 \Rightarrow A^2$ is invertible.

45. A

46. A

Sol. (for Q.45 & Q.46)

We have $(t_2 + t_3)(t_2 + t_1) = -4$

$\Rightarrow t_2^2 + t_2(t_1 + t_3) + t_1t_3 + 4 = 0$ … (1)

$\Rightarrow (t_1 + t_3)^2 - 4(t_1t_3 + 4) \geq 0 \Rightarrow (t_1 - t_3)^2 \geq 16$

$\Rightarrow t_2 + t_4 = -(t_1 + t_3)$ and $t_2t_4 = t_1t_3 + 4$(Because t_2 and t_4 are the roots of the equation (1))

also let y_1 and y_2 are y-coordinates of intersection points of the tangents at (A, C) and (B, D) then

$y_1 = t_1 + t_3$ and $y_2 = t_2 + t_4$ $\Rightarrow y_1 + y_2 = 0$.

47. C

48. C

Sol. (for Q.47 & Q.48)

(I) $x \geq 0 \Rightarrow \sqrt{x+9} \geq 3$

(II) $13 - x^2 = (x+1)^2 \Rightarrow x^2 + x - 6 = 0$

$\Rightarrow x = -3, 2$

(III) $x + 5 = 1 + x + 2\sqrt{x} \Rightarrow \sqrt{x} = 2 \Rightarrow x = 4$

(IV) Put $x - 1 = t^2$, to obtain

$|t+1| - |t-1| = 2$

which is satisfied for each $t \geq 1$

SECTION – C

49. 00000.60

Sol. $b + c = 4a$

$\sin B + \sin C = 4\sin A$

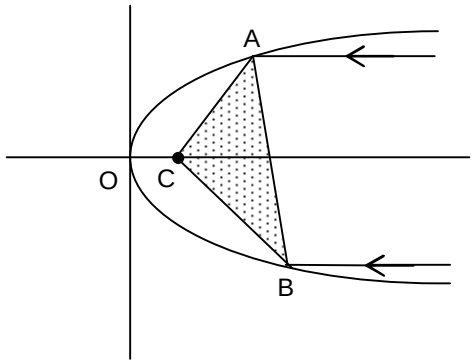
$\cos\frac{B-C}{2} = 4\cos\frac{B+C}{2}$

$\cos\frac{B}{2}\cos\frac{C}{2} + \sin\frac{B}{2}\sin\frac{C}{2} = 4\left[\cos\frac{B}{2}\cos\frac{C}{2} - \sin\frac{B}{2}\sin\frac{C}{2}\right]$

$3\cos\frac{B}{2}\cos\frac{C}{2} = 5\sin\frac{B}{2}\sin\frac{C}{2}$

50. 00002.62

Sol. Solving $y = 1$ and $y^2 = x$
then $A = (1, 1)$
and solving $y = -2$
and $y^2 = x$
then $B = (4, -2)$
$\because$ After reflection both reflected rays pass through focus of the parabola $y^2 = x$
i.e. $C = \left(\frac{1}{4}, 0\right)$



51. 00009.50

Sol. $\cos 4x = 2\cos^2 2x - 1$

$= 2\left(2\cos^2 x - 1\right)^2 - 1$

$= 2\left(4\cos^4 x + 1 - 4\cos^2 x\right) - 1$

$= 8\cos^4 x - 8\cos^2 x + 1$

$\therefore a_0 = 1, a_1 = -8, a_2 = 8$

$\therefore 5a_0 + a_1 + a_2 = 5$

52. 00000.50

Sol. $\lim\limits_{x\to 0} \dfrac{x\tan 2x - 2x\tan x}{4\sin^4 x}$

$= \lim\limits_{x\to 0} \dfrac{x}{4\sin^4 x}\left[\dfrac{2\tan x}{1-\tan^2 x} - 2\tan x\right]$

$= \lim\limits_{x\to 0} \dfrac{2x\tan x}{4\sin^4 x}\left[\dfrac{1-1+\tan^2 x}{1-\tan^2 x}\right]$

$= \lim\limits_{x\to 0} \dfrac{2x\tan^3 x}{4\sin^4 x\left(1-\tan^2 x\right)}$

$= \dfrac{1}{2}\lim\limits_{x\to 0} \dfrac{x}{\sin x}\ \dfrac{1}{\cos^3 x}\dfrac{1}{1-\tan^2 x}$

$= \dfrac{1}{2}.1.\dfrac{1}{1^3}.\dfrac{1}{1-0} = \dfrac{1}{2}$

53. 00000.20

Sol. $I = \left\{\sin^{-1}\left(\dfrac{x}{2\cos\alpha}\right)\right\}_0^{\sin 2\alpha} = \sin^{-1}\left(\dfrac{\sin 2\alpha}{2\cos\alpha}\right)$

$= \sin^{-1}\left(\sin\alpha\right)$

$= \pi - \alpha \quad \left(\because \pi < \alpha < \dfrac{2\pi}{3}\right)$

$\Rightarrow I = \pi - \alpha \qquad \text{or } \dfrac{I+\alpha}{\pi} = 1$

54. 00000.75

Sol. In ΔCAE, $AE^2 = AC^2 + CE^2 - 2AC.CE\cos C = \frac{73}{9}$

Also, in ΔCAE

$$\cos\theta = \frac{AC^2 + AE^2 + CE^2}{2.AC.AE} = \frac{8}{\sqrt{73}} \Rightarrow \tan\theta = \frac{3}{8}$$